PHYSICS III: General Physics 1

Heat and temperature

- Concept of heat and Internal energy
- Temperature and Zeroth Law of thermodynamics
- Thermodymic temperature Scale
- ~~Thermody~~ Thermometer (Types of thermometer and their Calculation).

Comcept of heat and internal energy

Heat is the net energy transferred from one body to another due to temperature difference. e.g a cold spoon in a hot coffee cup. When a body gains heat from a heat source, its internal molecular energy rises and subsequently increases its temperature heat added to a body is proportional to the body's temperature provided the boiling point is not reached. At boiling point, there is no longer temperature rise, rather the heat added breaks up the existing molecular bond.

Heat is measured in Joule(J). Other units are the calories

$1 \text{ cal} = 4.1868 J. = 3.968 \times 10^{-3}$

Conservation of energy

This states that the total amount of work expressed in any form (heat or work) is always the same.

This principles is expressed as:

$$Q = \Delta U + \Delta W$$

Q = amount of energy added to or removed from a body

ΔU = Change in internal energy of a body due to a given heat

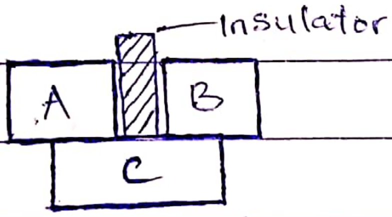
ΔW = Work done by or on the body.

Temperature and Zeroth Law of thermodynamics

- **Temperature:** This is defined as the relative measure of hotness and Coldness of a body. It is measured with a thermometer which Contains a working substance with a measurable property, Such as length or pressure that change in a regular way as the substance becomes hotter or Colder. It's S.I unit is the Kelvin (K).

- **Thermal equilibrium:** When two bodies (systems) are in thermal contact and there is no net flow of heat between them, then they are thermal equilibrium. Thus when a thermometer and some other objects are placed in contact with each other, they eventually reach thermal equilibrium. The reading of the thermometer is then taking to be the temperature of the other object. This is based on the Zeroth law of thermodynamics.

- **Zeroth Law of thermodynamics:** This states that if two bodies are each in ~~thermal~~ thermal equilibrium with a third body, then all three bodies are in thermal equilibrium with each other.



In other words, if C is initially in thermal equilibrium with A and B, then A and B are also in thermal equilibrium with each other.

Thermodynamics Temperature Scale

For a thermometer to become a useful measuring device for temperature, an appropriate scale is needed with numbers on it. For scientific measurement, the following scale are mentioned:

~~(1) Celcius Scale:~~ This uses the ~~boiling temperature of water at 100 (lower fixed points)~~ as a ~~fixed point on a scale with the~~

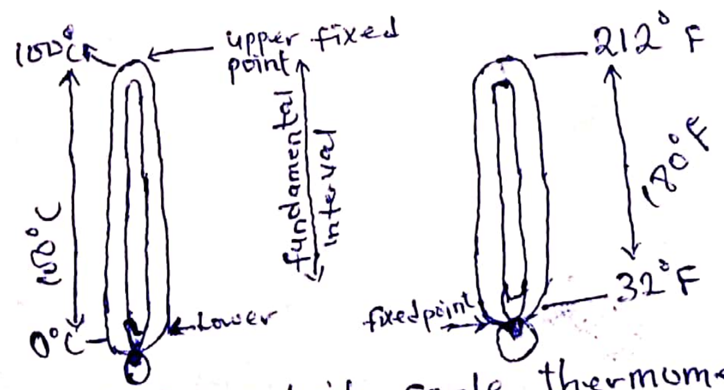
~~distance between~~ (1) **Celcius Scale**: This is the boiling temperature of water at 100 (upper fixed point) and the freezing temperature of water at 0 (lower fixed points) as a fixed points on a Scale. With the distance between each fixed point divided into 100 equal part, each called a degree and the distance between 0° to 100° called the fundamental interval.

(2) **Fahrenheit Scale**: Here, the lower fixed point (freezing temperature of H_2O) is ~~212°F as standard atmospheric pressure~~ 32°F and the upper fixed point (cooling temperature of H_2O) is 212°F as as standard atmospheric pressure. There are 180° between the upper and the lower fixed points.

(3) **Kelvin or Thermodynamics Temperature Scale or ideal gas Scale**:
This Scale uses one fixed point called the tripple point of water. This is the unique Combination of temperature and pressure at Solid water (ice), liquid water, and water vapour can coexist.

It occurs at temperature. $T_3 = 273.16K$, Corresponding to 0.01°c and a water vapour pressure of 610 Pa.

The Kelvin Scale is an absolute temperature Scale and it's zero point ($T = 0k = -273.15°c$), the temperature at which $p = 0$ is called absolute zero.



Celcius and fahrenheit scale thermometer

Temperature Conversion from one Scale to another

(1) $\frac{T_c - 0}{100} = \frac{T_F - 32}{180}$; $T_F = \frac{9}{5} T_c + 32^\circ$ — °c to °F

(2) $T_c = \frac{5}{9}(T_F - 32^\circ)$ ---------- °F to °c

(3) $T_K = T_c + 273.16$ ------ °c to K

Types of thermometer

(1) Liquid in ~~gas~~ glass thermometer e.g Clinical thermometers

It consist of capillary tube of uniform with bulb at one end, while the other end is sealed off. Mercury or alcohol fills the bulb and forms thread in the capillary tube of the thermometer.

Expansion of liquids, when exposed to heat results in increase in length of mercury or alcohol column L. If L_0 and L_{100} are the length of a mercury column at 0°c and 100°c and L_θ, the length at some other temperature θ, then θ in °c is defined by the equation.

$$\theta = \frac{L_\theta - L_0}{L_{100} - L_0} \times 100$$

Examples:

(1) In a certain thermometer (alcohol in glass) the 0°c mark corresponds to 1.88cm, the 100°c marks corresponds to 2.28cm. What will be the length of the alcohol corresponding to a temperature of 17°c

$L_0 = 1.88\text{cm}$, $L_{100} = 2.28\text{cm}$, $L_{17} = ?$ $\theta = 17°c$

$$17 = \frac{L_\theta - L_0}{L_{100} - L_0} = \frac{L_{17} - 1.88}{2.28 - 1.88} \times 100$$

$L_{17} = 1.948\text{cm}$

(2) A mercury thermometer reads 22.8mm in ice and 242mm in stem. It later reads 56.25mm on a summer day. What is the temperature of the day?

$L_0 = 22.8mm, \quad L_{100} = 242mm, \quad L_\theta = 56.25mm, \quad \theta = ?$

$$\theta = \frac{L_\theta - L_0}{L_{100} - L_0} \times 100 = \frac{56.25 - 22.8}{242 - 22.8} \times 100$$

$\theta = 15.3°c$

(2) **Resistance Thermometer:** Platinum is the material commonly used in this type of thermometer because of it high melting point and its inability to take part in chemical reactions

In a resistant thermometer the change in electrical resistance of a Coil of ~~the~~ fine wire with change in temperature is measured. Because resistance can be measured precisely, this thermometer are more precise than others.

The temperature θ corresponding to unknown resistance R_θ is calculated as: $\theta = \frac{R_\theta - R_0}{R_{100} - R_0} \times 100$

R_0 = Resistance at ice point

R_{100} = Resistance at steam point

R_θ = Resistance at an unknown temperature.

Example:

(1) The resistance of element of a platinum resistance thermometer is 2Ω at ice point and 2.7Ω at the steam point. What value of resistance would correspond to a temperature of 880°c?

$R_0 = 2\Omega, \quad R_{100} = 2.7\Omega, \quad \theta = 880°c, \quad R_\theta = ?$

$$\theta = \frac{R_\theta - R_0}{R_{100} - R_0} \times 100$$

$R_\theta = (0.7 \times 8.8) + 2 = 8.16\Omega$

(2) The resistance R_T of a resistance thermometer varies with temperature $T°c$ as measured by a gas thermometer according to the equation

$R_T = R_0(1 + 4000\beta T - \beta T^2)$. Calculate the temperature on ~~resistance~~ resistance scale corresponding to 200°C on this gas scale.

Solution: $\theta = \frac{R_\theta - R_0}{R_{100} - R_0} \times 100°C$

$R_T = R_{200} = R_0[1 + 4000(200)\beta - 200^2\beta]$

$R_{200} = R_0[1 + 760000\beta]$

$R_T = R_{100} = R_0[1 + 4000(100)\beta - 100^2\beta]$

$R_{100} = R_0[1 + 390000\beta]$

~~Substitu~~ Substituting the above equations

$$T = \frac{R_0[1 + 760000\beta] - R_0}{R_0[1 + 390000\beta] - R_0} \times 100$$

$$= \frac{R_0 + 760000\beta R_0 - R_0}{R_0 + 390000\beta R_0 - R_0} \times 100$$

$T = 194.9°c$

Gas Thermometers: These are themometers that do not depend some what on the specific properties of the material used. There are two types of gas thermometer:

(1) **Constant Volume gas thermometers:** It working principle is that the pressure of a gas at constant volume increases with temperature. If P_0, P_{100} and P_θ are the respective pressure at 0°c, 100°c and an unknown temperature θ, then θ in °C is defined by the equation

$$\theta = \frac{P_\theta - P_0}{P_{100} - P_0} \times 100°C$$

Example:

(1) A gas thermometer is used to measure the temperature of a furnace. The thermometer indicates the ice ~~steam~~ point and steam point as 40mmHg

and 70mmHg respectively. It records 50mmHg at an unknown temperature. What is the unknown temperature?

$P_0 = 40mmHg$ $P_{100} = 70mmHg$, $P_\theta = 50mmHg$ $\theta = ?$

$$\theta = \frac{P_\theta - P_0}{P_{100} - P_0} \times 100°C$$

$$= \frac{50 - 40}{70 - 40} \times 100°C$$

$$= 33.3°C$$

(2) A gas thermometer uses a gas of which the pressure varies with temperature according to the relation. $P = P_0(1 + BT - CT^2)$ where P_0, B and C are constant and T is the temperature in °C. Given that $P = 5mmHg$ at ice point, $P = 7mmHg$ at steam point and $P = 12mmHg$ at 210°C. Calculate P_0, B and C.

Solution: $P_0 = 5mmHg$, $P_{100} = 7mmHg$, $P_\theta = 12mmHg$ at 210°C

When $T = 0°C$; $P = 5mmHg$

$$\Rightarrow 5 = P_0[1 + B(0) - C(0)^2]$$

$$P_0 = 5mmHg$$

When $T = 210°C$, $P = 12mmHg$

$$\Rightarrow 12 = P_0[1 + B(210) - C(210)^2]$$

$$12 = 5[1 + 210B - 44100C]$$

$$2.4 = 1 + 210B - 44100C \quad ---- (i)$$

When $T = 100°C$, $P = 7mmHg$

$$\Rightarrow 7 = 5[1 + B(100) - C(100)^2]$$

$$0.4 = 1 + 100B - 10000C \quad ---- (ii)$$

Solving the ~~two~~ equations simultaneously i.e

$$2.4 = 1 + 210B - 44100C$$

$0.4 = 1 + 100B - 10000C$

It is found that

$C = 2.43 \times 10^{-5}$

$B = 1.56 \times 10^{-3}$

(2) **Constant Pressure gas thermometer:** At constant pressure, the volume of a gas increases with absolute temperature. If V_0, V_{100} and V_θ are the respective volumes at $0°C$, $100°C$ and an unknown temperature θ, then $\theta°C$ is defined ~~as~~ by the equation:

$$\theta = \frac{V_\theta - V_0}{V_{100} - V_0} \times 100°C$$

Example: A constant mass of gas maintained at a constant pressure has a volume of $2 \times 10^4 m^3$ at $0°C$, $2.73 \times 10^4 m^3$ at $100°C$ and $5.25 \times 10^4 m^3$ at normal boiling point of sulphur. Calculate the value of sulphure corresponding to the boiling point of sulphur.

$V_0 = 2 \times 10^4 m^3$, $V_{100} = 2.73 \times 10^4 m^3$, $V_\theta = 5.25 \times 10^4 m^3$

$$\theta = \frac{V_\theta - V_0}{V_{100} - V_0} \times 100°C$$

$$= \frac{(5.25 - 2) \times 10^4}{(2.73 - 2 \times 10^4} \times 100$$

$$\theta = 445.2°C$$

Thermocouple: Based on seeback's discovery, if two ends of wires of different metals are joined together, a small electric current flows when one junction is heated and the other one cooled. The common type of thermocouple thermometer uses bimetallic strip made by bonding strip of different metals together. When temperature of the composite strip increases, one metal expands more than the other and the strip bends. This strip is usually formed into spiral, with the outer end anchored to the thermometer case and the inner case end attached to a pointer.

The pointer rotates in response to temperature - changes. Temperature measured by thermocouple is defined by the relation:

$$E = a + bT + cT^2 \text{ ; } a, b, c \text{ are constant}$$

$$\theta = \frac{E_\theta - E_0}{E_{100} - E_0} \times 100°c$$

Example:

The e.m.f values of a thermocouple at temperature $0°c$; $\theta°c$ and $100°c$ are $E_0 = 0.75mV$, $E_\theta = 0.82mv$ and $E_{100} = 0.95mv$ respectively find $-53.8°c$ $E_{50} = 1.2611 mv$

(a) The value of temperature T in °c on the thermocouple scale

(b) The value of temperature on thermocouple scale corresponding to $50°c$ on the standard scale assumming that the thermocouple e.m.f at $T°c$ on the standard scale is given by, $E_T = E_0(1 + aT + 0.0008\,aT^2)$, a = constant

$E_0 = 0.75mv$, $E_\theta = 0.82mv$, $E_{100} = 0.95mv$

(a) $$\theta = \frac{E_\theta - E_0}{E_{100} - E_0} \times 100°c$$

$$= \frac{0.82 - 0.75}{0.95 - 0.75} \times 100°c$$

$$\theta = 35°c$$

(b) $$E_T = E_0(1 + aT + 0.008aT^2)$$

$$E_{50} = E_0\left[1 + 50a + 0.0008(50)^2 a\right]$$

$$= E_0\left[1 + \cancel{108}\,52a\right] \quad \text{---} \; (i)$$

again $$E_{100} = E_0\left[1 + 100a + 0.0008(100)^2 a\right]$$

$$= E_0\left[1 + 108a\right] \quad \text{---} \; (ii)$$

Now $$\theta = \frac{E_{50} - E_0}{E_{100} - E_0} \times 100°c$$

$$= \frac{E_0[1 + 52a] - E_0}{E_0[1 + 108a] - E_0} \times 100°c$$

$$\theta = \frac{529}{1089} \times 100°C$$

$$= 48.1°C$$

Optical Pyrometers: Some thermometers work by detecting the amount of infrared radiation emitted by an object (that all objects emits e.m.r, including IR as a consequence of their temperature) Pyrometers are used to measure very high temperature in ~~other~~ order of many hundreds to few thousands; Infrared from the hot body is read by the pyrometers detector until it's scale eventually stops at a number. This numbers corresponds to the temperature measured.

Temporal artery thermometer uses its IR sensor to measure the radiation from the temporal artery when run over the patients forehead. This gives more accurate value of body temperature than the oral or ear thermometers.